

Three Probes Suffice for Directional Localization on the Three-Cube

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September 9, 2026

Abstract

In the partial-feedback directional localization game, a probe that misses the robber returns an adversarially chosen first step of a shortest path toward the robber. Jones and Kinnersley proved that the directional localization number of the n -cube belongs to $\{n, n+1\}$ and asked for its exact value. We settle the three-dimensional instance: $\zeta_d(Q_3) = 3$. The first probe triple $(000, 000, 111)$ either locates the robber or leaves one of six two-vertex fibers. After the robber's possible move, a fiber-dependent second triple separates all remaining positions. Thus three cops win within two probing rounds. The construction is also checked by exact belief-state synthesis and an independently implemented exhaustive replay.

1 The game and the result

Jones and Kinnersley introduced the directional localization game and proved that

$$\zeta_d(Q_n) \in \{n, n+1\}$$

for every positive integer n ; they then asked for the exact value on hypercubes [1, Corollary 3.9 and Question 6.1]. We resolve only the first undecided slice, $n = 3$; no assertion about higher cubes is made.

We use the partial-feedback convention of that paper. Write $V(Q_3) = \mathbb{F}_2^3$, with adjacent vertices differing in one coordinate. If a probe is placed at p while the robber is at r , the set of legal replies is

$$D(p, r) = \begin{cases} \{p\}, & p = r, \\ \{p + e_j : p_j \neq r_j\}, & p \neq r, \end{cases}$$

where e_1, e_2, e_3 are the standard basis vectors and addition is in $\mathbb{F}_2^3$. Thus a missed probe returns one neighbor of p lying on a shortest p - r path, with the choice made adversarially. For a simultaneous probe tuple, each component reply may be chosen independently. The cops win as soon as the public replies determine the current vertex uniquely; otherwise the robber may stay or move along one edge before the next round. Repeated probes are allowed.

Equivalently, if B is the current public belief, $P = (p_1, p_2, p_3)$ is a probe tuple, and $A = (a_1, a_2, a_3)$ is the observed reply tuple, then its fiber is

$$B'(B, P, A) = \{r \in B : a_t \in D(p_t, r) \text{ for } t = 1, 2, 3\}.$$

A singleton fiber is an immediate win. Otherwise the next belief is the closed neighborhood $N[B']$, accounting for both staying and a one-edge move.

Theorem 1. *For the partial-feedback directional localization game on the three-cube,*

$$\zeta_d(Q_3) = 3.$$

Moreover, three cops can guarantee localization within two probing rounds.

2 An explicit two-round strategy

Put $\mathbf{1} = e_1 + e_2 + e_3$. The first-round probe tuple is

$$(0, 0, \mathbf{1}).$$

Lemma 2. *Every reply fiber for $(0, 0, \mathbf{1})$ is a singleton except for six fibers. For distinct $i, j \in \{1, 2, 3\}$, let k be the remaining coordinate. The reply*

$$(e_i, e_i, \mathbf{1} + e_k)$$

has precisely the fiber

$$F_{ij} = \{e_i, e_i + e_j\}.$$

Proof. A reply 0 to either zero probe is possible only when the robber is at 0, and a reply $\mathbf{1}$ to the last probe is possible only when the robber is at $\mathbf{1}$; either event locates immediately. For every other robber position r , a zero probe can return exactly an e_i with $i \in \text{supp}(r)$, while the $\mathbf{1}$ probe can return exactly an $\mathbf{1} + e_k$ with $k \notin \text{supp}(r)$.

Suppose the two zero-probe replies are e_i and e_j . If $i \neq j$, then $i, j \in \text{supp}(r)$ and the last reply identifies the absent third coordinate, so $r = e_i + e_j$. If both replies are e_i , the last reply identifies an absent coordinate $k \neq i$. With j denoting the remaining coordinate, the support of r is either $\{i\}$ or $\{i, j\}$, giving exactly F_{ij} . Both positions in F_{ij} admit the displayed reply, so the list is exact. $\square$

Assume that the ambiguous fiber F_{ij} occurs. After the robber stays or moves one edge, the next belief is

$$B_{ij} = N[F_{ij}] = \{0, e_i, e_j, e_i + e_j, e_i + e_k, \mathbf{1}\}. \quad (1)$$

In the second round the cops probe

$$(0, e_i + e_j, e_i + e_k). \quad (2)$$

To verify separation, abbreviate $a = e_i$, $b = e_j$, and $c = e_k$. For each candidate in (1), the three component sets of legal replies to the probes in (2) are as follows.

robber r	$D(0, r)$	$D(a + b, r)$	$D(a + c, r)$
0	$\{0\}$	$\{a, b\}$	$\{a, c\}$
a	$\{a\}$	$\{a\}$	$\{a\}$
b	$\{b\}$	$\{b\}$	$\{a, c, a + b + c\}$
$a + b$	$\{a, b\}$	$\{a + b\}$	$\{a, a + b + c\}$
$a + c$	$\{a, c\}$	$\{a, a + b + c\}$	$\{a + c\}$
$a + b + c$	$\{a, b, c\}$	$\{a + b + c\}$	$\{a + b + c\}$

For every pair of distinct rows, at least one component set is disjoint. The Cartesian products of possible reply triples are therefore pairwise disjoint. Every legal second-round reply identifies the current robber vertex before any further movement. Together with Lemma 2, this proves $\zeta_d(Q_3) \leq 3$. The matching lower bound follows from Jones–Kinnersley [1, Corollary 3.9], which gives $\zeta_d(Q_3) \geq 3$. This proves Theorem 1.

3 Exact computational check

The preceding argument is the proof, but the finite game was also checked in two separate implementations. A bit-mask retrograde solver enumerates all 255 nonempty beliefs and all $8^3 = 512$ ordered three-probe tuples, including tuples with repetitions. It assigns rank 0 to the eight singleton beliefs, rank 1 to all 246 non-singleton proper beliefs, and rank 2 to the full belief. Its selected rank-2 action is $(0, 0, 7)$ in binary encoding, and its only non-singleton reply fibers are the six in Lemma 2.

A standalone replay checker does not import the solver. It represents cube vertices as coordinate triples, reconstructs adjacency and shortest-path replies from their definitions, and verifies strict rank decrease after every unresolved certified reply. It checks all 255 certified beliefs and 3472 legal reply fibers; the six unresolved transitions all go from rank 2 to rank 1. From the project root the calculation can be reproduced with

```
python3 evidence/solve_q3.py
python3 evidence/check_q3_certificate.py
```

The resulting finite certificate is corroborating evidence and is not used as a substitute for the explicit proof.

Scope and assistance statement

The result concerns only the partial-feedback game on Q_3 , with simultaneous probes, independently adversarial legal replies, and a stay-or-one-edge robber move after an unresolved round. It does not settle the value on Q_n for $n > 3$. The literature comparison is limited to the cited primary source, and no exhaustive priority claim is made.

OpenAI Codex assisted with exact search, execution of the two verification programs, proof organization, and manuscript preparation. A separate model context rederived the reply fibers and replayed the certificate; this was an internal verification pass, not peer review. The deterministic scripts and their machine-readable outputs accompany the article, while the displayed two-round strategy provides the mathematical proof.

References

- [1] John Jones and William B. Kinnersley. The directional localization game on graphs, 2026. arXiv:2609.01745v1.